INCH-POUND

FF-B-575D
19 MAY 2022
SUPERSEDING
FF-B-575C
26 MAY 1970

DETAIL SPECIFICATION SHEET

Bolt, Hexagon and Square

Inactive for new design after 4 December 2000
and is no longer used, except for replacement purposes.

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

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AMSC N/A FSC 5306

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1. SCOPE

1.1. Scope. This specification covers the physical and mechanical properties of various types of hexagon and square head bolts having nominal diameters of ¼ inch and larger.

1.2. Classification. Bolts shall be of the following types and sizes, as specified (see 6.2).

1.2.1. Types.

Type I – Square head bolts.
Type II – Hexagon head bolts.
Type III – Heavy hexagon head bolts.
Type IV – Heavy hexagon head structural bolts.

1.2.2. Sizes. Bolts shall be classified according to size by the basic thread diameter.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents.

DEPARTMENT OF DEFENSE STANDARDS

QQ-N-281 – Nickel-Copper Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections.

QQ-N-286 – Nickle-Copper-Aluminum Alloy, Wrought.

MIL-F-495 – Finish, Chemical, Black, for Copper Alloys.

MIL-PRF-8625 – Anodic Coatings, for Aluminum and Aluminum Alloys.

MIL-DTL-16232 – Phosphate Coating, Heavy, Manganese or Zinc Base.

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product.

MS35355 – Bolt, Machine, Square Head (Regular), Steel, Cadmium or Zinc Plated, UNC-2A

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MS35356 – Bolt, Machine, Square Head (Unfinished), Steel, Phosphate Coated, UNC-2A.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ).

ASQ H1331 – Zinc-Nickel Electroplating for Fasteners

(Copies of this document are available for purchase online from www.asq.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS2700 – Passivation of Corrosion Resistant Steels

SAE AMS-QQ-P-416 – (R) Plating, Cadmium (Electrodeposited)

SAE AIR4127 – Steel: Chemical Composition and Hardenability

SAE AMS-QQ-A-225/6 – Aluminum Alloy Bar, Rod and Wire; Rolled, Drawn or Cold Finished, 2024.

SAE J429 – Mechanical and Quality Requirements for Externally Threaded Fasteners.

(Copies of these documents are available for purchase online from www.sae.org.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) STANDARD.

ASME B18.2.1 – Square and Hex Bolts and Screws.

ASME B1.1 – Unified Inch Screw Threads (UN and UNR Thread Form)

(Copies of this document are available for purchase online from www.asme.org.)

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M – Zinc Coating (Hot-Dip) On Iron and Steel Hardware.

ASTM F3125/F3125M – Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength.

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ASTM A370 – Mechanical Testing of Steel Products.

ASTM E8/E8M – Standard Test Methods for Tension Testing of Metallic Materials

ASTM E10 – Brinell Hardness of Metallic Materials.

ASTM E18 – Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E112 – Estimating the Average Grain Size of Metals.

ASTM B21/B21M – Standard Specification for Naval Brass Rod, Bar, and Shapes.

ASTM B138/B138M – Standard Specification for Manganese Bronze Rod, Bar, and Shapes.

ASTM B139/B139M - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes.

ASTM B98/B98M - Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes.

ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.

ASTM A342/A342M – Standard Test Methods for Permeability of Weakly Magnetic Materials.

ASTM E1282 – Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

(Copies of these documents are available for purchase online from www.astm.org.)

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIERMENTS

3.1. Material. Unless otherwise specified (see 6.2), bolts shall be made of the materials specified in 3.1.1 thru 3.1.3.

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3.1.1. Carbon and alloy steel. Carbon and alloy steel shall be in accordance with Grades 1, 2, 5, 7 or 8 of SAE J429, except for Type IV – Structural Bolts, which shall be in accordance with ASTM F3125/F3125M, as specified (see 6.2).

3.1.2. Corrosion-resisting steel. Corrosion-resisting steel shall be in accordance with the SAE 30300 (UNS S3000) series of SAE AIR4127 or any other austenitic corrosion-resisting steel developed for cold heading or free machining, i.e., steel which meets the mechanical properties specified herein.

3.1.2.1. Magnetically permeability. Corrosion-resisting steel bolts shall have a magnetic permeability of 2.0 max (air = 1.0) for a field strength of H = 200 oersteds when using a magnetic indicator in accordance with ASTM A342/A342M.

3.1.3. Non-ferrous. Non-ferrous bolts shall be made from the materials specified in Table III.

3.2. Mechanical properties.

3.2.1. Carbon and alloy steel. Mechanical properties of carbon and alloy steel bolts shall be in accordance with Table II and Table IV.

3.2.2. Corrosion-resisting steel. Mechanical properties of nickel-chromium and chromium alloy steel bolts shall be in accordance with the following for diameters thru 5/8 inch:

a.	Ultimate tensile strength	80,000 p.s.i. min.
b.	Yield strength (0.2% offset)	30,000 p.s.i. min.
c.	Rockwell hardness	B80 min.
d.	Elongation	30 percent min.
e.	Reduction in area	40 percent min

For diameters over .625 inch, ultimate tensile strength shall be 70,000 p.s.i. min and have a Rockwell hardness of B74 min.

3.2.3. Non-ferrous. Mechanical properties of non-ferrous bolts shall be in accordance with Table III.

3.3. Protective finish.

3.3.1. Cadmium plating. When specified (see 6.2), carbon and alloy steel bolts shall be cadmium plated in accordance with SAE AMS-QQ-P-416, Type II, class 3. Cadmium is an R1, R2 and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, type K16 is the preferred alternative.

3.3.2. Zinc-Nickel Plating. When specified (see 6.2), carbon and alloy steel bolts shall be Zinc-Nickel plated in accordance with MIL-PRF-32647, Type K16. This is the preferred alternative to Cadmium plating.

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3.3.3. Zinc coating. When specified (see 6.2), carbon and alloy steel bolts shall be zinc coated in accordance with ASTM B633, FeZn5, Type VI.

3.3.3.1. Zinc coating (hot dip). When specified (see 6.2), carbon and alloy steel bolts shall be hot dip galvanized in accordance with ASTM A153/A153M.

3.3.4. Phosphate coating. When specified (see 6.2), carbon and alloy steel bolts shall be phosphate coated in accordance with MIL-DTL-16232, type z, class 2.

3.3.5. Hydrogen embrittlement. Alloy steel bolts required to be electroplated or phosphate coated shall be subjected to a hydrogen embrittlement relief treatment conducted in accordance with the applicable plating or coating specification, immediately after plating or coating to minimize

3.3.6. Passivation. Unless otherwise specified (see 6.2), corrosion-resisting steel bolts shall be passivated in accordance with SAE AMS2700.

3.3.7. Anodic coating. Unless otherwise specified (see 6.2), aluminum alloy bolts shall be anodized in accordance with MIL-PRF-8625, type II, class 1.

3.3.8. Black oxide. When specified (see 6.2), brass bolts shall be treated with a black chemical finish in accordance with MIL-F-495.

3.4. Dimensions. Dimensions and tolerances of bolts shall be in accordance with MS35355, MS35356 and ASME B18.2.1, as specified (see 6.2).

3.4.1. Lengths. Unless otherwise specified (see 6.2), bolts shall be furnished in the following length increments:

Lengths thru 5/8 inch	- 1/16 inch increments
Over 5/8 thru 1-1/2 inches	- 1/8 inch increments
Over 1-1/2 thru 5 inches	- 1/4 inch increments
Over 5 inches	- 1/2 inch increments

3.4.2. Threads. Bolt threads shall be Class 2A, UNC, UNF, UNRC or UNRF, as specified (see 6.2), in accordance with ASME B1.1 except that the UNR threads shall have a root radius between the limits of 0.10825p and 0.14434p. Thread dimensions shall not exceed the basic size after plating or coating.

3.5. Metallographic requirements.

3.5.1. Decarburization. When specifies (see 6.2), the partial decarburization of threads of carbon and alloy steel bolts shall not exceed the amounts shown in Figure 1.

3.5.2. Grain size of copper alloy. When specified (see 6.2), the grain size of copper alloy bolts shall be determined in accordance with the applicable material specification.

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3.6. Surface discontinuities (Type IV – Structural bolts only). Type IV bolts shall contain no longitudinal discontinuities (located parallel to the axis of the bolt in the threads, body, fillet, or underside of head) with a depth normal to the surface greater than 0.03D, where D is the nominal bolt size in inches. Type IV bolts shall contain no transverse cracks (located perpendicular to the axis of the bolt in the threads, body, fillet or underside of head). Type IV bolts shall contain no bursts having a width greater than 0.010 in. plus 0.025D, where D is the nominal bolt size in inches. This requirement applies to bolts made of ASTM F3125/F3125M Grade A490 material only.

3.7. Head marking. Bolts shall be marked with the applicable grade identification symbol in accordance with SAE J429, ASTM F3125/F3125M. Bolts shall also be marked with the manufacturer's identification symbol. Markings shall be located on the top of the head, and shall be either raised or depressed, at the manufacturer's option.

3.8. Workmanship. Bolts shall be free from burrs, rust, nicks, lumps and other surface contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Inspection lot. A lot shall consist of all bolts of the same type, size, material, finish, threads and length, produced by the same manufacturer under essentially the same conditions, and offered for acceptance at one time.

4.3. Sampling for lot acceptance.

4.3.1. Sampling for examination. Unless otherwise specified in the contract (see 6.2), for sampling for examination, an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. Acceptable verification levels for MIL-STD-1916 and index values for ASQ H1331 shall be specified in Table I.

4.3.2. Sampling for test of mechanical properties, metallographic requirements and chemical analysis. A random sample of bolts shall be taken from each lot in accordance with MIL-STD-1916, verification level III or ASQ H1331 with an index value of 1.5.

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4.3.3. Sampling for test of discontinuities. Sampling for test of discontinuities shall be in accordance with ASTM F3125/F3125M. (Type IV – Structural bolts made of ASTM F3125/F3125M Grade A490 material only.)

4.3.4. Sampling for protective finish. Sampling for tests of protective finish shall be in accordance with the applicable specification of 3.3.

4.3.5. Sampling for packaging and packing. Sampling for test of preservation, packaging, and marking shall be specified in the contract order (see 5.1)

4.4. Examination. Each bolt taken as specified in 4.3.1 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table I. Any bolt having one or more defects shall be rejected and if the number of defective bolts in any sample exceeds the acceptance number of the sample, the represented lot shall be rejected.

Table I. Classification of Defects

Categories	Defects	Inspection Method
Critical	None defined	
Major	Verification level III, Index Value = 2.5	
101	Threads, not as specified (3.4.2)	SIE*
102	Type and size, not as specified (1.2)	SIE
103	Length, not as specified (3.4.1)	SIE
104	Width across flats, not as specified (3.4)	SIE
Minor	Verification level II, Index Value = 4.0	
201	Other dimensions, not as specified (3.4)	SIE
202	Magnetic permeability (3.1.2.1)	SIE
203	Head marking (3.7)	Visual
204	Workmanship (3.8)	Visual

* Standard Inspection Equipment

4.5. Tests.

4.5.1. Hardness. Samples taken in accordance with 4.3.2 shall be hardness tested in accordance with ASTM E10 or ASTM E18 to verify conformance with requirements of this specification. A transverse section through the threaded portion of the bolts shall be taken one nominal diameter from the point, or when too short, halfway between the bearing surface and the point. The hardness shall be measured at a point on this section, one quarter diameter from the axis of the bolt.

4.5.2. Proof load. Samples taken in accordance with 4.3.2 shall be proof load tested in accordance with ASTM A370 – Supplement III to verify conformance with the requirements of

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this specification. Bolts too large to be proof load tested full size shall be tested using machine test specimens (see 4.5.3.2).

4.5.3. Tensile strength.

4.5.3.1.Full size bolts. Samples taken in accordance with 4.3.2 shall be tensile strength tested in accordance with ASTM E8/E8M to verify conformance with the requirements of this specification.

4.5.3.2.Machines test specimens. When bolts cannot be tested full size for proof load and tensile strength requirements, tests shall be conducted using test specimens machined from the bolts.

For 1.5 diameter and larger bolts a standard 0.500 in. round, 2.000 in. gage length shall be turned from the bolt with the axis of the specimen located midway between the center and outside surface of the bolt, as shown in figure 3. Bolts 0.75 thru 1.375 in. diameter shall have their shanks machined to a standard 0.500 in. round, 2.000 in. gage length test specimen concentric with the axis of the bolt, leaving the bolt head and threaded section intact, as shown in figure 2. Bolts 0.25 thru 0.625 in. diameter shall have their shanks machined to a standard 0.250 in. round, 1.000 in. gage length test specimen, as shown in figure 2.

The test specimens shall be proof load tested in accordance with 4.5.2 and tensile strength tested in accordance with 4.5.3.1.

4.5.3.2.1. When specified (see 6.2), machined test specimens shall be tested for yield strength, elongation and reduction in area in accordance with ASTM A370 – supplement III.

4.5.3.3. Wedge tensile strength. Samples taken in accordance with 4.3.2 shall be tensile strength tested with a wedge in accordance with ASTM A370 – Supplement III, to verify conformance with the requirements of this specification. Bolts used in the proof load test may be used for this test.

4.5.3.4. Short bolts. Bolts too short to be tensile strength tested shall be subjected to a hardness test in accordance with 4.5.1 to verify conformance with the requirements of this specification.

4.5.4. Metallographic tests.

4.5.4.1.Decarburization. when specified (see 6.2), samples taken in accordance with 4.3.2 shall be subjected to a decarburization test. The threaded part shall be sectioned longitudinally through the axis and polished so that rounding of the thread edges is held to a minimum. The section shall be etched with the usual metallographic etchants (such as nital) and examined with a metallographic microscope. Measurements shall be made from the crest of the thread to the end of the decarburized zone. Such distance shall not exceed that indicated in Figure 1. Measurements shall be made with a fixed scale micrometer eye-piece, or the image of the

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threading may be thrown on the ground glass of the camera and measured with dividers and a scale. This test does not apply to corrosion-resisting steel.

4.5.4.2.Grain size of copper alloys. When specified (see 6.2), copper alloy samples taken in accordance with 4.3.2 shall be tested for grain size in accordance with ASTM E112.

4.5.5. Chemical analysis. When specified (see 6.2), samples taken in accordance with 4.3.2 shall be tested for chemical analysis in accordance with ASTM E1282. When a material certification can be furnished, it will be acceptable.

4.5.6. Discontinuities. Samples taken in accordance with 4.3.2 shall be tested for discontinuities in accordance with ASTM F3125/F3125M, to verify conformance to 3.6.

4.5.7. Protective finish. Examination and test of protective finish shall be in accordance with the applicable specification of 3.3.

4.5.8. Packaging and packing. Examination and test of preservation, packaging and marking shall be specified in the contract order (see 5.1).

4.5.9. Hydrogen embrittlement. The contractor shall furnish the Government certification that electroplated or phosphate coated alloy bolts have been subjected to the hydrogen embrittlement relief treatment specified in 3.3.4. When specified (see 6.2), electroplated or phosphate coated alloy steel bolts shall be subjected to the embrittlement relief test specified in the applicable plating or coating specification, except that the bolts shall be held under load for 23 hours and then examined for cracks or fractures.

5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1. Intended use. Hexagon and square head bolts are designed for insertion through holes in assembled parts and are normally intended to be tightened or released by torquing a nut.

6.2. Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title number and date of this specification.

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b. Type and size required (1.2).
c. Material (3.1).
d. Protective finish required (3.3).
e. Length required (3.4.1).
f. Threads (3.4.2).
g. Other tests, if required (4.5.3.2.1).
h. Metallographic tests, if required (4.5.4).
i. Chemical analysis, if required (4.5.5).
j. Embrittlement relief test, if required (4.5.9).
k. level of preservation, packing and packaging required (5.1).

6.3. Civil procurement. When level B packaging or packing is required for civil agency procurement, the requirements of level A will apply (see 5.1).

6.4. Supersession data. This specification includes the requirements of FF-B-571a in part, dated January 9, 1934 and FF-B-575b, dated September 12, 1960. The cross reference between the types is as follows:

FF-B-571a		FF-B-575b		FF-B-575c	
Type A1 –	Square head, unfinished bolts	Type 1 –	Regular square bolt	Type I –	Square head bolts
Type A2 –	Square head, semi-finished bolts		None		None
Type A3 –	Square head, finished bolts		None		None
Type B1 –	Hexagonal head, Unfinished bolts	Type 2 –	Regular hexagon bolt	Type II –	Hexagon head bolts
Type B2 –	Hexagonal head, semi-finished bolts	Type 3 –	Regular semifinished hexagon bolts		None
Type B3 –	Hexagonal head, finished bolts				none
		Type 4 –	Heavy hexagon bolts	Type III –	Heavy hexagon head bolts
		Type 5 –	Heavy semi-finished hexagon bolts		None
		Type 6 –	Heavy finished hexagon bolts		None
				Type IV –	Heavy hexagon structural bolts.

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6.5. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

CONCLUDING MATERIAL

Custodians:
Army – AR
Air Force – 99
DLA – GS

Preparing Activity:
DLA – GS7

(Project 5306-2022-014)

Review Interests:
Army – AT, AV, CE, EA, MI
Navy – MC, OS, YD

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Table II. Mechanical Properties of Steel Bolts

GRADE	NOMINAL SIZE	FULL SIZE BOLTS		MACHINED TEST SPECIMENS			HARDNESS			
		PROOF LOAD PSI	TENSILE STRENGTH MIN, PSI	YIELD STRENGTH AT 0.2% OFFSET MIN, PSI	ELONGA-TION IN 2.00 IN. MIN-%	REDUC-TION IN AREA MIN-%	BRINELL		ROCKWELL	
							MIN	MAX	MIN	MAX
1	All Sizes	33,000	60,000	36,000	18	35	121	241	B70	B100
2	.25 thru .75 *	55,000	74,000	57,000	18	35	149	241	B80	B100
	Over .75 thru 1.5**	33,000	60,000	36,000	18	35	121	241	B70	B100
5	.25 thru 1	85,000	120,000	92,000	14	35	255	321	C25	C34
	Over 1 thru 1.5	74,000	105,000	81,000	14	35	223	285	C19	C30
	Over 1.5 thru 3	55,000	90,000	58,000	14	35	183	235	B90	C22
7	.25 thru 1.5	105,000	133,000	115,000	12	35	277	321	C28	C34
8	.25 thru 1.5	120,000	150,000	130,000	12	35	302	352	C32	C38
	Over 1.5 thru 3	85,000	120,000	92,000	12	35	255	321	C25	C34
ASTM F3125/F3125M GRADE A325	All Sizes	85,000	120,000	92,000	14	35	253	319	C25	C34
ASTM F3125/F3125M GRADE A490	All Sizes	120,000	150,000	130,000	14	40	311	352	C33	C38

* 6” and less in length.
** All diameters if length exceeds 6”.

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Table III. Mechanical Properties of Non-Ferrous Bolts

MATERIAL	APPLICABLE SPECIFICATION	COMP. OR CLASS	CONDITION	TENSILE STRENGTH P.S.I. MIN.	YIELD STRENGTH AT 0.2 PERCENT OFFSET OR AT EXTENSION INDICATED		ELONGATION PERCENT MIN.
					P.S.I. MIN.	EXTENSION UNDER LOAD	
Manganese bronze	ASTM B138/B138M	UNS C67500	Soft-anneal	55,000	22,000	0.5 percent	20 3/
Phosphor bronze	ASTM B139/B139M	UNS C51000	Hard	60,000	33,000	---	15 1/
Silicon Bronze	ASTM B98/B98M	UNS C65100	Hard	60,000	40,000	0.5 percent	10 3/
Naval brass	ASTM B21/B21M	UNS C46400	Half-hard	60,000	27,000	0.5 percent	25 3/
Aluminum alloy	AMS-QQ-A-225/6	2024	Temper T4	62,000	40,000	0.0058 in. per in.	10 3/
Nickel-copper alloy	QQ-N-281	Class A	Hot finished	80,000	40,000	0.0071 inch per 2 inches	30 1/
Nickel-copper-aluminum alloy	QQ-N-286	Class A	Age hardened	130,000	90,000	---	20 1/

1/ In 2 inch gauge length
2/ In 4 times diameter or thickness (min)
3/ In 2 inches or 4 times diameter (min)

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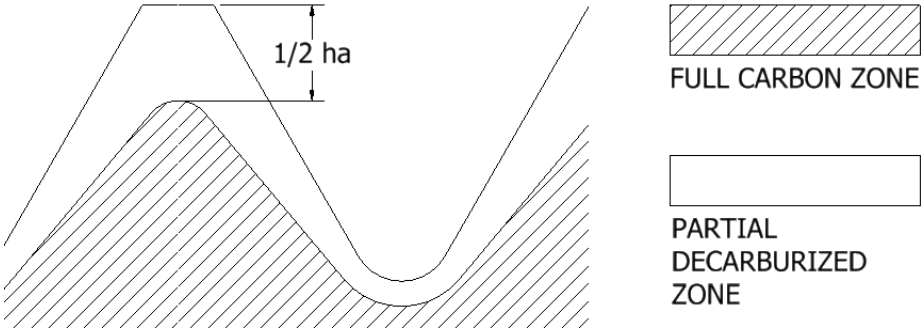
Table IV. Proof Load and Tensile Strength Requirements for Steel Bolts

NOMINAL SIZE AND THREAD	STRESS AREA SQ. IN.	GRADE 1		GRADE 2		GRADE 5		GRADE 7		GRADE 8		ASTM F3125/F3125M GRADE A325		ASTM F3125/F3125M GRADE A490	
		PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN.	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN	PROOF LOAD LBS	TENSILE STRENGTH LBS-MIN
COARSE THREAD SERIES – UNC															
¼-20	.0318	1,050	1,900	1,750	2,350	2,700	3,800	3,350	4,250	3,800	4,750	---	---	---	---
5/16-18	.0524	1,750	3,150	2,900	3,900	4,450	6,300	5,500	6,950	6,300	7,850	---	---	---	---
3/8-16	.0775	2,550	4,650	4,250	5,750	6,600	9,300	8,150	10,300	9,300	11,600	---	---	---	---
7/16-14	.1063	3,500	6,400	5,850	7,850	9,050	12,800	11,200	14,100	10,800	15,900	---	---	---	---
½-13	.1419	4,700	8,500	7,800	10,500	12,100	17,000	14,900	18,900	17,000	21,300	12,050	17,050	17,050	21,300
9/16-12	.182	6,000	10,900	10,000	13,500	15,500	21,800	19,100	24,200	21,800	27,300	---	---	---	---
5/8-11	.226	7,450	13,600	12,400	16,700	19,200	27,100	23,700	30,100	27,100	33,900	19,200	27,100	27,100	33,900
¾-10	.334	11,000	20,000	18,400	24,700	28,400	40,100	35,100	44,400	40,100	50,100	28,400	40,100	40,100	50,100
7/8-9	.462	15,200	27,700	15,200	27,700	39,300	55,400	48,500	61,400	55,400	69,300	39,250	55,450	55,450	69,300
1-8	.606	20,000	36,400	20,000	36,400	51,500	72,700	63,600	80,600	72,700	90,900	51,500	72,700	72,700	90,900
1-1/8-7	.763	25,200	45,800	25,200	45,800	56,500	80,100	80,100	101,500	91,600	114,400	64,900	91,600	91,550	114,450
1-1/4-7	.969	32,000	58,100	32,000	58,100	71,700	101,700	101,700	127,700	116,300	145,400	82,400	116,300	116,300	145,350
1-3/8-6	1.155	38,100	69,300	38,100	69,300	85,500	121,300	121,300	153,600	138,600	173,200	98,200	138,600	138,600	173,250
1-1/2-6	1.405	46,400	84,300	46,400	84,300	104,000	147,500	147,500	186,900	168,600	210,800	119,500	168,600	168,600	210,750
FINE THREAD SERIES -UNF															
¼-28	.0364	1,200	2,200	2,000	2,700	3,100	4,350	3,800	4,850	4,350	5,450				
5/16-24	.0580	1,900	3,500	3,200	4,300	4,900	6,950	6,100	7,700	6,950	8,700				
3/8-24	.0878	2,900	5,250	4,800	6,500	7,450	10,500	9,200	11,700	10,500	13,200				
7/16-20	.1187	3,900	7,100	6,550	8,800	10,100	14,200	12,500	15,800	14,200	17,800				
½-20	.1599	5,300	9,600	8,800	11,800	13,600	19,200	16,800	21,300	19,200	24,000				
9/16-18	.203	6,700	12,200	11,200	15,000	17,300	24,400	21,300	27,000	24,400	30,400				
5/8-18	.256	8,450	15,400	14,100	18,900	21,800	30,700	26,900	34,000	30,700	38,400				
¾-16	.373	12,300	22,400	20,500	27,600	31,700	44,800	39,200	49,600	44,800	56,000				
7/8-14	.509	16,800	30,500	16,800	30,500	43,300	61,100	53,400	67,700	61,100	76,400				
1-12	.663	21,900	39,800	21,900	39,800	56,400	79,600	69,600	88,200	79,600	99,400				
1-1/8-12	.856	28,200	51,400	28,200	51,400	63,300	89,900	89,900	113,800	102,700	128,400				
1-1/4-12	1.073	35,400	64,400	35,400	64,400	79,400	112,700	112,700	142,700	128,800	161,000				
1-3/8-12	1.315	43,400	78,900	43,400	78,900	97,300	138,100	138,100	174,900	157,800	197,200				
1-1/2-12	1.581	52,200	94,900	52,200	94,900	117,000	166,000	166,000	210,300	189,700	237,200				

NOTES:

1. Proof loads and tensile strengths are computed by multiplying the proof load stresses and tensile strength stresses given in table II by the stress area of the thread.
2. The stress area of sizes and thread series not included in this Table may be computed from the formula $A_s = 0.7845\left[D - \frac{0.9743}{n}\right]^2$ Where D = nominal diameter in inches, n=threads per inch.

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THREADS PER INCH	1/2 ha		THREADS PER INCH	1/2 ha
80	.002		18	.009
72	.002		16	.010
64	.003		14	.012
56	.003		13	.012
48	.003		12	.014
44	.004		11	.015
40	.004		10	.016
36	.005		9	.018
32	.005		8	.020
28	.006		7	.023
24	.007		6	.027
20	.008			

1/2 ha = 1/2 ADDENDUM

Figure 1. Decarburization Classification of Bolt Threads

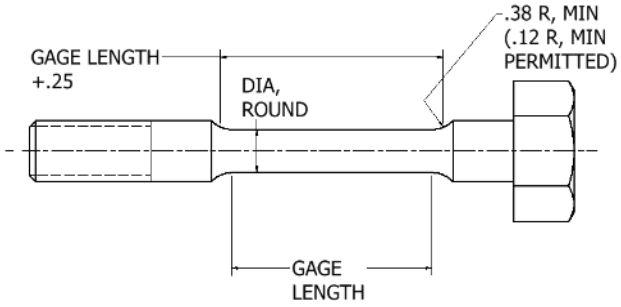


Figure 2. Test Specimen for Bolts with Turned Down Shank

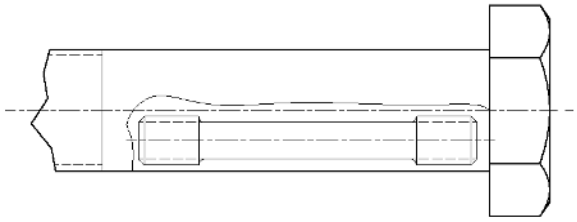


Figure 3. Location of Standard Round 2.00 Inch Gage Length Test Specimen When Turned from Large Size Bolt

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